Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the heat load of a control enclosure, its natural temperature rise and the airflow for a target rise. It also gives the fan duty point and the cooling-unit capacity needed when the limit cannot be met.

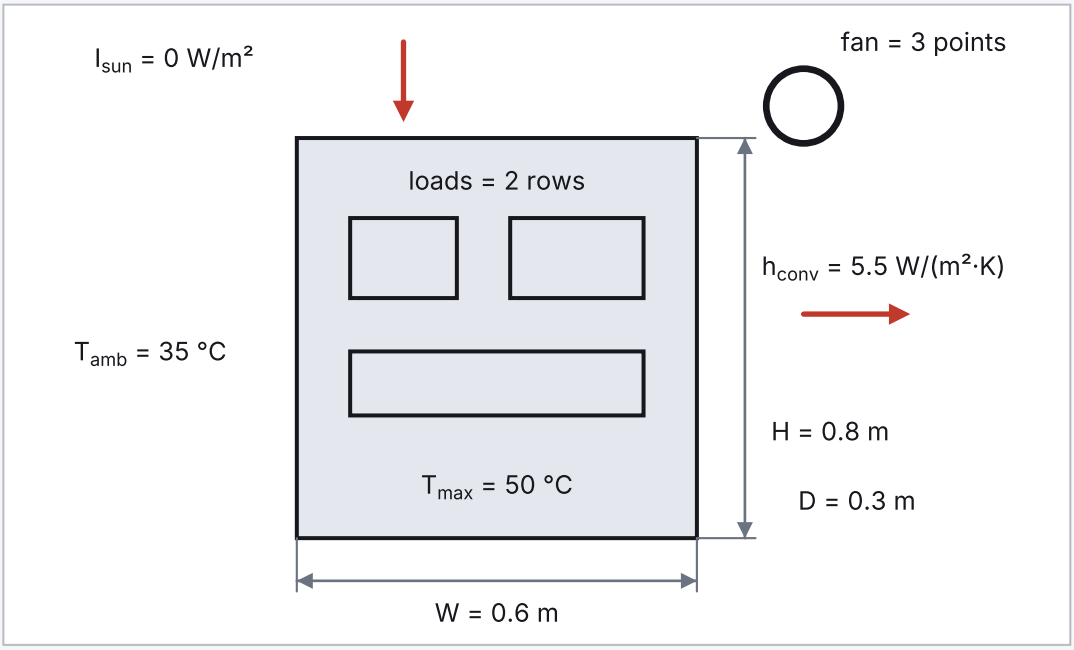


Figure 1 — An enclosure W by H by D holding the heat loads as listed, losing heat through its exposed faces to the ambient, with a fan and the sun on it. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Heat dissipated inside the enclosure, plus any solar gain	$P = \Sigma P_i n_i d_i (+ \alpha I A_{sun})$
Effective cooling surface, each face weighted by how freely it sheds heat	$A_{eff} = \Sigma f_{face} \times A_{face}$
Temperature rise with natural cooling, one coefficient over the exposed faces (the IEC TR 60890 tables are not used)	$\Delta T = P/(h_{conv} \cdot A_{eff})$
Air flow a fan must move to hold the target rise	$V = P/(\rho\, c_p\, \Delta T_{target})$, with $\rho = p(z)/(287.055\ T)$ and c_p = 1006 J/(kg·K)
Fan duty point, where the fan curve meets the system resistance (found by bisection)	$p_{fan}(V) = p_{sys}\,(V/V_{ref})^2$
Cooling unit capacity, when the surfaces alone cannot reject the heat	$P_{cool} = P - h_{conv} \cdot A_{eff} \cdot (T_{max} - T_{amb})$; with T_{max} at or below T_{amb} the term is a gain, $P_{cool} = P + h_{conv} \cdot A_{eff} \cdot (T_{amb} - T_{max})$, and no ventilation airflow or fan duty is given

WHAT IT CALCULATES	RELATION
Sealed enclosure (sealing temperature T_{seal} entered, no breather)	$\Delta p = p_{site} \cdot \lvert T_{int}/T_{seal} - 1 \rvert$, the pressure difference across the walls when the air warms or cools from the sealing temperature at constant volume, in absolute temperatures (a rise when $T_{int} > T_{seal}$, a suction when below); a breather or relief element limits it, $dp_{seal} = \min(\Delta p, dp_{relief})$; the force on a cover of area A_{cover} is $F_{seal} = dp_{seal} \cdot A_{cover}$

Basis: A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation[1]; Verification of temperature rise (edition to be checked)[2](cl. 10.10).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Heat loads	2 row(s)		spec: Panel schedule PNL-04	any
W	Enclosure width	0.6	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
H	Enclosure height	0.8	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
D	Enclosure depth	0.3	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
f_{top}	Top exposure (1 free, 0 covered)	1		assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1
f_{front}	Front exposure	1		assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1
f_{rear}	Rear exposure (0 = wall-mounted)	0		assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1
f_{left}	Left side exposure	1		assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{right}	Right side exposure	1		assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1
f_{bottom}	Bottom exposure (0 = on the floor)	0		assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1
A_{sun}	Sunlit area	0	m²	assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 100 m²
α_{sun}	Surface absorptivity	0.5		assumption: Not used indoors (no sun); template default kept.	0.1 … 1
h_{conv}	Overall heat-transfer coefficient of the enclosure surface (editable reference value)	5.5	W/(m²·K)	reference: Reference record: painted steel enclosure, still air, 5.5 W/m²K (manufacturer guide)	3 … 8 W/(m²·K)
P_{add}	Additional heat load (linked from the load-budget dissipated heat or another calculation)	0	kW	assumption: No further heat load linked into this panel.	0 … 100 kW
I_{sun}	Solar irradiance on the enclosure (0 = indoors)	0	W/m²	assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 1200 W/m²
T_{amb}	Ambient temperature	35	°C	spec: Site conditions SC-2: 35 °C plant room; 50 °C inside limit for the drives	-20 … 55 °C
altitude	Site altitude (air density)	0	m	assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 4000 m
T_{max}	Maximum internal temperature	50	°C	spec: Site conditions SC-2: 35 °C plant room; 50 °C inside limit for the drives	20 … 80 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_{sys}	System resistance (filters, grilles) at the reference airflow	20	Pa	spec: Filter datasheet: 20 Pa at 100 m³/h	0 … 500 Pa
V_{ref}	Reference airflow for the system resistance	100	m³/h	spec: Filter datasheet: 20 Pa at 100 m³/h	1 … 100000 m³/h
V_{pts}	Fan curve: airflow points	[0, 100, 200]	m³/h	spec: Fan datasheet FF-120: curve points	any
p_{pts}	Fan curve: static pressure points	[60, 30, 0]	Pa	spec: Fan datasheet FF-120: curve points	any

TABLE ROW	VALUES
loads[1]	name = PLC and I/O; P = 80 W; qty = 1; duty = 1
loads[2]	name = drives; P = 55 W; qty = 4; duty = 1

3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h_{conv}. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h_{conv} is a reference value the user may edit.)
A2. The coefficient covers natural convection and radiation to surroundings at the ambient temperature. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h_{conv} is a reference value the user may edit.)
A3. Covered faces dissipate nothing. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h_{conv} is a reference value the user may edit.)
A4. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise. (Manufacturer fan-sizing practice.)
A5. Natural dissipation is neglected for the airflow, which is conservative. (Manufacturer fan-sizing practice.)
A6. A sealed enclosure (T_{seal} entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed. (Ideal gas at constant volume (derived).)
A7. The pressure difference follows the ideal-gas relation at constant volume, from the sealing temperature to T_{int}. (Ideal gas at constant volume (derived).)
A8. T_{int} is the internal temperature with natural dissipation, without a cooling unit. (Ideal gas at constant volume (derived).)
A9. A breather, a seal that leaks or a flexing wall reduces the difference, so the result is an upper bound. (Ideal gas at constant volume (derived).)
A10. A pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance. (Ideal gas at constant volume (derived).)
A11. The site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A12. The fan curve is linear between the entered points. (Duty-point idealisation.)
A13. The system resistance scales with airflow squared. (Duty-point idealisation.)
A14. f_{top} (Top exposure (1 free, 0 covered)) is taken as 1. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A15. f_{front} (Front exposure) is taken as 1. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A16. f_{rear} (Rear exposure (0 = wall-mounted)) is taken as 0. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A17. f_{left} (Left side exposure) is taken as 1. (Wall mounted on a floor plinth: rear and bottom covered;

indoors.)
A18. f_{right} (Right side exposure) is taken as 1. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A19. f_{bottom} (Bottom exposure (0 = on the floor)) is taken as 0. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A20. A_{sun} (Sunlit area) is taken as 0 m². (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A21. α_{sun} (Surface absorptivity) is taken as 0.5. (Not used indoors (no sun); template default kept.)
A22. P_{add} (Additional heat load (linked from the load-budget dissipated heat or another calculation)) is taken as 0 kW. (No further heat load linked into this panel.)
A23. I_{sun} (Solar irradiance on the enclosure (0 = indoors)) is taken as 0 W/m². (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A24. altitude (Site altitude (air density)) is taken as 0 m. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)
A25. The effective surface is the plain exposed-area method of manufacturer practice: each face area weighted by how freely it sheds heat, one overall coefficient h. This is not a verification of temperature rise to IEC TR 60890 or BS EN 61439-1 cl. 10.10 (neither the surface factors nor the standard's temperature-rise correlation is used). (Stated on every run so the result is not read as an assembly verification.)

4. Working

[18] $P_{loads} = \Sigma\ P_i \times n_i \times d_i$
$= 2\ \text{item(s)}$
$= 0.3\ \text{kW}$
Internal heat load

[19] $P_{solar} = \alpha_{sun} * I_{sun} * A_{sun}$
$= 0.5 \times 0\ \text{W/m}^2 \times 0\ \text{m}^2$
$= 0\ \text{kW}$
Solar gain

[20] $P_{total} = P_{loads} + P_{add} + P_{solar}$
$= 0.3\ \text{kW} + 0\ \text{kW} + 0\ \text{kW}$
$= 0.3\ \text{kW}$
Total heat to remove

[21] $A_{eff} = f_{top} * W * D + f_{bottom} * W * D + f_{front} * W * H + f_{rear} * W * H + f_{left} * H * D + f_{right} * H * D$ [1](verification)
$= 1 \times 0.6\ \text{m} \times 0.3\ \text{m} + 0 \times 0.6\ \text{m} \times 0.3\ \text{m} + 1 \times 0.6\ \text{m} \times 0.8\ \text{m} + 0 \times 0.6\ \text{m} \times 0.8\ \text{m} + 1 \times 0.8\ \text{m} \times 0.3\ \text{m} + 1 \times 0.8\ \text{m} \times 0.3\ \text{m}$
$= 1.14\ \text{m}^2$
Effective dissipating surface

[22] $dT_{nat} = P_{total} / (h_{conv} * A_{eff})$
$= 0.3\ \text{kW} / (5.5\ \text{W/(m}^2\cdot\text{K)} \times 1.14\ \text{m}^2)$
$= 47.847\ \text{K}$
Temperature rise with natural dissipation

[23] $T_{int} = T_{amb} + dT_{nat}$
$= 35\ \text{degC} + 47.847\ \text{K}$
$= 82.847\ \text{degC}$
Internal temperature with natural dissipation

[24] $dT_{allow} = T_{max} - T_{amb}$
$= 50\ \text{degC} - 35\ \text{degC}$
$= 15\ \text{K}$
Allowed rise

[25] $u_{nat} = dT_{nat} / dT_{allow}$
$= 47.847\ \text{K} / 15\ \text{K}$
$= 3.1898$
Natural dissipation utilisation

[26] $P_{nat,max} = h_{conv} * A_{eff} * dT_{allow}$
= 5.5 W/(m²·K) × 1.14 m² × 15 K
= 0.09405 kW
Heat the enclosure dissipates naturally at the limit
[27] $P_{cool} = max(0\ W, P_{total} - P_{nat,max})$
= max(0 W, 0.3 kW − 0.09405 kW)
= 0.20595 kW
Cooling-unit capacity needed if ventilation is not used
[28] $p_{site} = 101325\ (1 - 2.25577e\text{-}5\ z)^{5.2559}$
= z = 0 m
= 101.32 kPa
Site pressure
[29] $\rho_{air} = p_{site} / (287.055\ J/(kg*K) * T_{amb})$
= 101.32 kPa / (287.055 J/(kg*K) × 35 degC)
= 1.1455 kg/m³
Air density at ambient and altitude
[30] $V_{req} = P_{total} / (\rho_{air} * 1006\ J/(kg*K) * dT_{allow})$
= 0.3 kW / (1.1455 kg/m³ × 1006 J/(kg*K) × 15 K)
= 62.481 m³/h
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[31] $V_{req,net} = max(0\ m^3/h, (P_{total} - P_{nat,max}) / (\rho_{air} * 1006\ J/(kg*K) * dT_{allow}))$
= max(0 m^3/h, (0.3 kW − 0.09405 kW) / (1.1455 kg/m³ × 1006 J/(kg*K) × 15 K))
= 42.893 m³/h
Airflow crediting natural dissipation
[36] $V_{duty} = p_{fan}(V) = p_{sys}\ (V/V_{ref})^2$, bisection
= $p_{fan}(V) = 20 × (V/100)^2$, V from 0 to 200 m³/h
= 113.75 m³/h
Fan duty airflow
[37] $p_{duty} = p_{sys}\ (V_{duty}/V_{ref})^2$
= 20 × (113.7/100)²
= 25.876 Pa
Static pressure at duty
[38] $u_{fan} = V_{req} / V_{duty}$
= 62.481 m³/h / 113.75 m³/h
= 0.5493
Airflow utilisation (required / delivered)

5. Results

SYMBOL	VALUE	DESCRIPTION
P_{total}	0.3000 kW	Total heat
P_{cool}	0.2060 kW	Cooling unit needed
$P_{nat,max}$	0.09405 kW	Natural dissipation at the limit
T_{max}	50 °C	Internal temperature limit
dT_{nat}	47.85 K	Natural temperature rise (governing)
T_{int}	82.85 °C	Internal temperature (natural)
V_{req}	62.48 m³/h	Required airflow
V_{duty}	113.7 m³/h	Fan duty airflow
P_{loads}	0.3000 kW	Internal heat load

SYMBOL	VALUE	DESCRIPTION
P_{solar}	0 kW	Solar gain
A_{eff}	1.140 m²	Effective surface
ρ_{air}	1.145 kg/m³	Air density
$V_{req,net}$	42.89 m³/h	Required airflow (net)
p_{duty}	25.88 Pa	Fan duty pressure
u_{fan}	0.55	Airflow utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Natural temperature rise against the allowed rise	47.85 K	15 K	3.19	
PASS	Required airflow against the fan duty airflow	62.48 m³/h	113.7 m³/h	0.55	

7. Applicability

WARNING T_{max}: natural dissipation holds 94.05 W of 300 W. Ventilate, or fit a 206 W cooling unit. [cooling-unit]

8. Verification

Template elec.enclosure-heat@1.7.1 Content hash c419a03dc9212448 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. International Electrotechnical Commission. A Method of Temperature-Rise Verification of Low-Voltage Switchgear and Controlgear Assemblies by Calculation. IEC TR 60890. IEC.
2. British Standards Institution. Low-Voltage Switchgear and Controlgear Assemblies—Part 1: General Rules. BS EN 61439-1. BSI.